-- Funções trigonométricas

tangente : $tg(x) = \frac{\operatorname{sen} x}{\cos x}$, $D = \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\}$

cotangente : $\operatorname{cotg}(x) = \frac{\cos x}{\operatorname{sen} x}$, $D = \mathbb{R} \setminus \{k\pi : k \in \mathbb{Z}\}$

Secante : $\sec(x) = \frac{1}{\cos x}$, $D = \mathbb{R} \setminus \{\frac{\pi}{2} + k\pi : k \in \mathbb{Z}\}$

Cossecante : $\operatorname{cosec}(x) = \frac{1}{\operatorname{sen}(x)}$, $D = \mathbb{R} \setminus \{k\pi : k \in \mathbb{Z}\}$

-- Funções trigonométricas inversas

Arco-seno : $\operatorname{arcsen}(x) = \operatorname{sen}^{-1}(x)$, $D = [-\pi/2, \pi/2]$

Arco-cosseno : $\arccos(x) = \cos^{-1}(x)$, $D = [0, \pi]$

Arco-tangente : $\operatorname{arctg}(x) = tg^{-1}(x)$, $D =]-\pi/2, \pi/2[$

Arco-cotangente : $\operatorname{arcotg}(x) = \operatorname{cotg}^{-1}(x)$, $D =]0, \pi[$

Arco-secante : $\operatorname{arcsec}(x) = \sec^{-1}(x)$, $D = [0, \pi/2[$

Arco-cossecante : $\operatorname{arcosec}(x) = \operatorname{cosec}^{-1}(x)$, $D =]0, \pi/2]$

-- Funções hiperbólicas

Seno hiperbólico : $sh(x) = \frac{e^x - e^{-x}}{2}$, $D = \mathbb{R}$

Cosseno hiperbólico : $ch(x) = \frac{e^x + e^{-x}}{2}$, $D = \mathbb{R}$

tangente hiperbólica : $th(x) = \frac{sh(x)}{ch(x)}$, $D = \mathbb{R}$

Cotangente hiperbólica: $\coth(x) = \dfrac{1}{th(x)}$, $D = \mathbb{R} \setminus \{0\}$

Secante hiperbólica: $sech(x) = \dfrac{1}{ch(x)}$, $D = \mathbb{R}$

Cossecante hiperbólica: $cosech(x) = \dfrac{1}{sh(x)}$, $D = \mathbb{R} \setminus \{0\}$

-- Funções hiperbólicas inversas

Argumento do seno hiperbólico:

$$argsh(x) = (sh)^{-1}(x), \quad D = \mathbb{R}$$

Argumento do cosseno hiperbólico:

$$argch(x) = (ch)^{-1}(x), \quad D = \mathbb{R}_0^+$$

Argumento da tangente hiperbólica:

$$argth(x) = (th)^{-1}(x), \quad D = \mathbb{R}$$

Argumento da cotangente hiperbólica:

$$argcoth(x) = (coth)^{-1}(x), \quad D = \mathbb{R} \setminus \{0\}$$

Argumento da secante hiperbólica:

$$argsech(x) = (sec)^{-1}(x), \quad D = \mathbb{R}_0^+$$

Argumento da cossecante hiperbólica:

$$argcosech(x) = (cosech)^{-1}(x), \quad D = \mathbb{R} \setminus \{0\}$$